Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[british]babel

[a4paper,top=2cm,bottom=2cm,left=2.5cm,right=2.5cm,marginparwidth=1.75cm]geometry

[style=apa, backend=biber]biblatex references.bib

britishbritish-apa [article]volume1

amsmath graphicx [colorlinks=true, allcolors=blue]hyperref hyperref [title]appendix mathrsfs amsfonts booktabs caption threeparttable algorithm algorithmicx algpseudocode listings enumitem chngcntr booktabs lipsum subcaption authblk [T1]fontenc csquotes diagbox comment

subcaption [justification=raggedright, singlelinecheck=false]caption caption helvet lmodern
setspace
titlesec

0.1em
lineno
float caption [table]position=top
Article title article title article title

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Ringkasan

Abstracts must be able to stand alone and so cannot contain citations to the paper's references, equations, etc. An abstract must consist of a single paragraph and be concise. Because of online formatting, abstracts must appear as plain as possible. Three to six keywords must be included. Each keyword should not exceed three words.

Keywords: keyword1, keyword2, keyword3, keyword4, keyword5, keyword6.

Nomenclature

T	Temperature (K)
u_i	Velocity in the x-direction (m/s)
τ_{ij}	Shear stress (N/m2)
ω	Specific turbulent dissipation rate (1/s)
Y_ω	Dissipation of ω

1 Introduction

Your introduction goes here! Simply start writing your document and use the Recompile button to view the updated PDF preview. Examples of commonly used commands and features are listed below to help you get started. [2]

Once familiar with the editor, you can find various project settings in the Overleaf menu, accessed via the button at the top left of the editor. To view tutorials, user guides, and further documentation, please visit our help library, or head to our plans page to choose your plan.

This is an example of a new paragraph with a numbered footnote[1] and a second footnote marker.[2]

[1] `https://data.gov.uk/`
[2] Example of footnote text.

2 Example of First Level Head - Section Head

[4]
[4]

2.1 How to create sections and subsections

Simply use the section and subsection commands, as in this example document! With Overleaf, all the formatting and numbering is handled automatically according to the template you've chosen. If you're using the Visual Editor, you can also create new sections and subsections via the buttons in the editor toolbar.

2.2 This is an example of second level head - subsection head

[5]

2.2.1 This is an example of third level head - subsubsection head

[6]

This is an example of fourth level head - paragraph head
[7]

3 Example of First Level Head

3.1 This is an example of second level head - subsection head

3.1.1 This is an example of third level head - subsubsection head

[8]

This is an example of fourth level head - paragraph head
[9]

4 How to Include Equations

Equations in LaTeX can either be inline or set as display equations. For inline equations use the `$...$` commands. Eg: the equation $H\psi = E\psi$ is written via the command `$H \psi = E \psi$`.

For display equations (with auto generated equation numbers) one can use the equation or eqnarray environments:

$$\|\tilde{X}(k)\|^2 \leq \frac{\sum\limits_{i=1}^{p} \left\|\tilde{Y}_i(k)\right\|^2 + \sum\limits_{j=1}^{q} \left\|\tilde{Z}_j(k)\right\|^2}{p+q}, \tag{1}$$

where $\mathrm{D}_\mu = \partial_\mu - ig\frac{\lambda^a}{2}A^a_\mu$
$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + gf^{abc}A^b_\mu A^a_\nu$ Notice the use of `\nonumber` in the align environment at the end of each line, except the last, so as not to produce equation numbers on lines where no equation

numbers are required. The `\label{}` command should only be used at the last line of an align environment where `\nonumber` is not used.

$$Y_\infty = \left(\frac{m}{\mathrm{GeV}}\right)^{-3}\left[1+\frac{3\ln(m/\mathrm{GeV})}{15}+\frac{\ln(c_2/5)}{15}\right] \tag{2}$$

The class file also supports the use of `\mathbb{}`, `\mathscr{}` and `\mathcal{}` commands. As such `\mathbb{R}`, `\mathscr{R}` and `\mathcal{R}` produces R, R and $\mathcal{R}$ respectively

Equations must be provided as editable text, either in a Word or LaTeX source file. They should be numbered consecutively through the manuscript as shown in Equations 1, 4 and 2. In APA style, when discussing numbered equations in the text, write out the word "Equation" and give the number. For example, you would write "see Equation 1." Use no punctuation after the equation if it appears at the end of a sentence; however, it is permissible (and may even be necessary) to place some form of punctuation after it (a comma or semi-colon, for example) if it appears in the middle of the sentence and is followed by text. In any case, maintain the coherence of all sentences with equations in them.

5 How to Include Tables

Use the table and tabular environments for basic tables — see Tables 1 and 2, for example. Table 1 is an sample figure including table footnotes. For more information, please see this help article on tables.

Tabel 1: Sample table with footnotes

column 1	column 2	column 3	column 4
row 1	data 1	data 2	data 3
row 2	data 4	data 51	data 6
row 3	data 7	data 8	data 92

Source: This is an example of table footnote. This is an example of table footnote. This is an example of table footnote. This is an example of table footnote. This is an example of table footnote.

Example of a first table footnote.

Example of a second table footnote.

6 How to Include Figures

First you have to upload the image file from your computer using the upload link in the file-tree menu. Then use the includegraphics command to include it in your document. Use the figure environment and the caption command to add a number and a caption to your figure. See the code for Figure 1 in this section for an example. As shown in Figures 1 and 2, the images should be single-page documents.

Note that your figure will automatically be placed in the most appropriate place for it, given the surrounding text and taking into account other figures or tables that may be close by. You can

Tabel 2: Example of a lengthy table which is set to full textwidth.

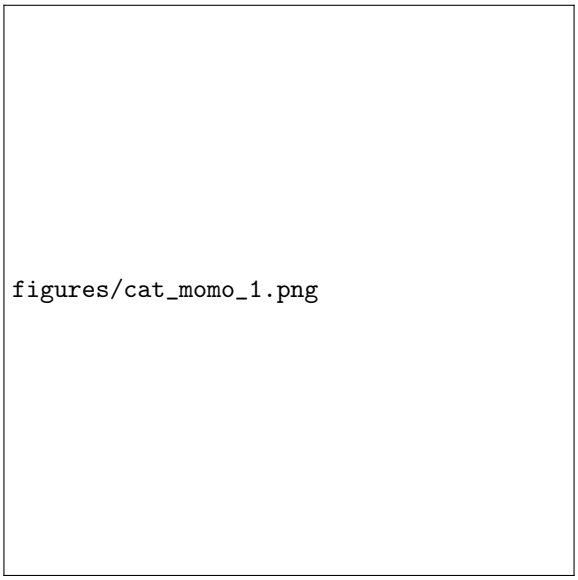
	Element 11			Element 22		
(lr)2-4(lr)5-7 Project	Energy	σ_{calc}	σ_{expt}	Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 ± 12	780 A	1166	1239 ± 100
Element 4	500 A	961	922 ± 10	900 A	1268	1092 ± 40

Note: This is an example of table footnote this is an example of table footnote this is an example of table footnote this is an example of table footnote this is an example of table footnote.

Example of a first table footnote.

Example of a second table footnote.

find out more about adding images to your documents in this help article on including images on Overleaf.



Gambar 1: This cat picture is located at the 'figures' folder.

If the figure contains multiple sub-figures, label each sub-figure at the top-left using the Helvetica bold font with a capital letter (e.g., A, B, C, etc.). When citing figures with multiple parts within the text, use the format "Figure 2A" and "Figure 2B". For the figure caption, employ the format "(A)", "(B)", etc.

6.1 More information about figures

As per display LaTeX standards one has to use eps images for `latex` compilation and `pdf/jpg/png` images for `pdflatex` compilation. This is one of the major differences between `latex` and `pdflatex`. The images should be single-page documents. The command for inserting images for `latex` and

pdflatex can be generalized. The package used to insert images in latex/pdflatex is the graphicx package. Figures can be inserted via the normal figure environment as shown in the below example:

```
\begin{figure}[h]
        \centering\includegraphics{<eps-file>}
        \caption{<figure-caption>}
        \label{<figure-label>}
\end{figure}
```

7 How to Include Algorithms, Program Codes, and Listings

Packages algorithm, algorithmicx, and algpseudocode are used for setting algorithms in latex. For this, one has to use the below format:

```
\begin{algorithm}
\caption{<alg-caption>}\label{<alg-label>}
\begin{algorithmic}[1]
. . .
\end{algorithmic}
\end{algorithm}
```

You may need to refer to the above-listed package documentation for more details before setting an algorithm environment. To set program codes, one has to use the program package. We need to use the \begin{program} ... \end{program} environment to set program codes.

[!ht] Calculate $y = x^n$ [1] $n \geq 0 \vee x \neq 0$ $y = x^n$ $y \Leftarrow 1$ $n < 0$ $X \Leftarrow 1/x$ $N \Leftarrow -n$ $X \Leftarrow x$ $N \Leftarrow n$ $N \neq 0$ N is even $X \Leftarrow X \times X$ $N \Leftarrow N/2$ [N is odd] $y \Leftarrow y \times X$ $N \Leftarrow N - 1$

Similarly, for listings, one has to use the listings package. To set environments similar to the verbatim environment, the \begin{lstlisting} ... \end{lstlisting} environment is used . Refer to the lstlisting package documentation for more details on this.

language=Pascal[frame=single,framexleftmargin=-1pt,framexrightmargin=-17pt,framesep=12pt,linewidth=0.95] for i:=maxint to 0 do begin do nothing end; Write('Case insensitive '); Write('Pascal keywords.');

8 How to Include Lists

List in LaTeX can be of three types: numbered, bulleted, and unnumbered. The "enumerate" environment produces a numbered list, the "itemize" environment produces a bulleted list, and the "unlist" environment produces an unnumbered list. In each environment, a new entry is added via the \item command.

[label=0.]This is the 1st item Enumerate creates numbered lists, itemize creates bulleted lists, and unnumerate creates unnumbered lists.[label=.]

2. (a) Second level numbered list. Enumerate creates numbered lists, itemize creates bulleted lists, and description creates unnumbered lists.

(b) Second level numbered list. Enumerate creates numbered lists, itemize creates bulleted lists, and description creates unnumbered lists.
[label=()]Third level numbered list. Enumerate creates numbered lists, itemize creates bulleted lists, and description creates unnumbered lists. Third level numbered list. Enumerate creates numbered lists.

(ii) Second level numbered list. Enumerate creates numbered lists, itemize creates bulleted lists, and description creates unnumbered lists.

3. Numbered lists continue.

Lists in LaTeX can be of three types: enumerate, itemize, and description. In each environment, a new entry is added via the `\item` command.

- First level bulleted list. This is the 1st item
- First level bulleted list. Itemize creates bulleted lists, and description creates unnumbered lists.
  - Second level dashed list. Itemize creates bulleted lists, and description creates unnumbered lists.
  - Second level dashed list. Itemize creates bulleted lists, and description creates unnumbered lists.
- First level bulleted list. Bullet lists continue.

Example of unnumbered list items:

Sample unnumberd list text. Sample unnumberd list text. Sample unnumberd list text. Sample unnumberd list text. Sample unnumberd list text.

Sample unnumberd list text. Sample unnumberd list text. Sample unnumberd list text.

Sample unnumberd list text. Sample unnumberd list text. Sample unnumberd list text. Sample unnumberd list text.

9 How to Add Citations and a References List

You can simply upload a `.bib` file containing your BibTeX entries, created with a tool such as JabRef. You can then cite entries from it, like this: greenwade93. Just remember to specify a bibliography style, as well as the filename of the `.bib`. You can find a video tutorial here to learn more about BibTeX.

Here is an example citation when you want an author name like collins2011a to appear in the text. And here's how to do a parenthetic citation, when you want to mention a reference at the end of a sentence or part of a sentence collins2013. It is possible to cite multiple references at the same time collins2011b,collins2016,lunn2007a,lunn2007b,ross2006,shannon1948.

If you have an upgraded account, you can also import your Mendeley or Zotero library directly as a `.bib` file, via the upload menu in the file-tree.

9.1 Citation in text

Please ensure that every reference cited in the text is also present in the reference list (and vice versa). Citations in the text should follow the referencing style used by the American Psychological Association. You are referred to the Publication Manual of the American Psychological Association (APA), Seventh Edition, ISBN 978-1-4338-3215-4, copies of which may be ordered online. References in the Abstract should be avoided, but if essential, then cite the author(s) and year(s). Unpublished results and personal communications are not recommended in the reference list but may be mentioned in the text. If these references are included in the reference list, they should follow the standard reference style of the journal and should include a substitution of the publication date with either 'Unpublished results' or 'Personal communication'. The citation of a reference as 'in press' implies that the item has been accepted for publication.

An APA in-text citation includes only three items: the last name(s) of the author(s), the year the source was published, and sometimes the page or location of the information. More than one reference from the same author(s) in the same year must be identified by the letters 'a', 'b', 'c', etc., placed after the year of publication. The following paragraph shows examples of APA style of citations.

Here is an example citation when you want an author name like collins2011a to appear in the text. And here's how to do a parenthetic citation when you want to mention a reference at the end of a sentence or part of a sentence collins2013. It is possible to cite multiple references at the same time collins2011b,collins2016,lunn2007a,lunn2007b,ross2006,shannon1948.

The followings are examples of `\textcite{...}`: rahman2019centroidb, krizhevsky2012imagenet, horvath2018dna, and lecun2015deep, zhang2018fine, ravi2016deep. Another example of `\parencite{...}`: bahdanau2014neural,imboden2018cardiorespiratory,motiian2017unified,murphy2012machine,ji20123d.

9.2 References

The Reference Section, also called the Reference List or Cited Works List, is a list of the full-text details of the in-text citations that have been used in the main text. It includes information such as the name of the author(s), the year the source was published, the full title of the source, and the URL or page range. The Reference Section allows the reader to find the text easily and can be considered as the long-hand format of the in-text citation. It is found at the end of the piece of writing. The works in a reference section should be arranged first alphabetically and then further sorted chronologically if necessary.

9.2.1 Web references

As a minimum, the full URL and the date when the reference was last accessed should be given. Any further information, if known (DOI, author names, dates, reference to a source publication, etc.), should also be given. Web references can be listed separately (e.g., after the reference list) under a different heading if desired or can be included in the reference list. With standard numerical .bst files, only numerical citations are possible. With an author-year .bst file, both numerical and author-year citations are possible.

9.2.2 Examples of reference style

You can find information about the examples of APA-style references to various sources at the following site:
https://apastyle.apa.org/style-grammar-guidelines/references/examples.

10 Conclusions

Some conclusions here.

Conflicts of Interest

The authors must declare conflicts of interest or state "The authors declare no conflict of interest." Authors must identify and declare any personal circumstances or interests that may be perceived as inappropriately influencing the representation or interpretation of reported research results. A detailed definition of conflicts of interest is available at the following site: https://academic.oup.com/journals/pages/authors/preparing_your_manuscript/ethics#conflict.

Author Contributions

The authors must specify the individual contributions of all authors, identified by full names, according to NISO CrediT (Contributer Roles Taxonomy) described at the following site: https://credit.niso.org/. An example statement is as follows:
Kunwoo Lee: Conceptualization, Methodology, Software. **Shuming Gao**: Data curation, Writing—original draft. **Sang Hun Lee**: Visualization, Investigation. **Jami J. Shah**: Supervision. **Hiromasa Suzuki**: Software, Validation. **Myung-Il Roh**: Writing—review & editing.

Funding

Cite all funding for your research, providing the grant number and the funder name. An example statement is as follows: This work is supported in part by funds from the National Science Foundation (NSF: # 1636933 and # 1920920).

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Acknowledgments

The authors thank those people or institutions that have helped you in the preparation of the manuscript.

Appendix

11 Some Notation

[10]

11.1 Appendix subsection title here

As shown in Equation K1, the section number is inserted in the equation number. [11]

$$Y_\infty = \left(\frac{m}{\mathrm{GeV}}\right)^{-3} \left[1 + \frac{3\ln(m/\mathrm{GeV})}{15} + \frac{\ln(c_2/5)}{15}\right] \tag{K1}$$

11.2 Appendix subsection title here

As shown in Table K1, the section number is inserted in the table number. [12]

Tabel K1: Sample table with three parts and five columns

column 1	column 2	column 3	column 4	column 5
row 1	data 0	data 1	data 2	data 3
row 2	data 4	data 5	data 6	data 7
row 3	data 8	data 9	data 10	data 11

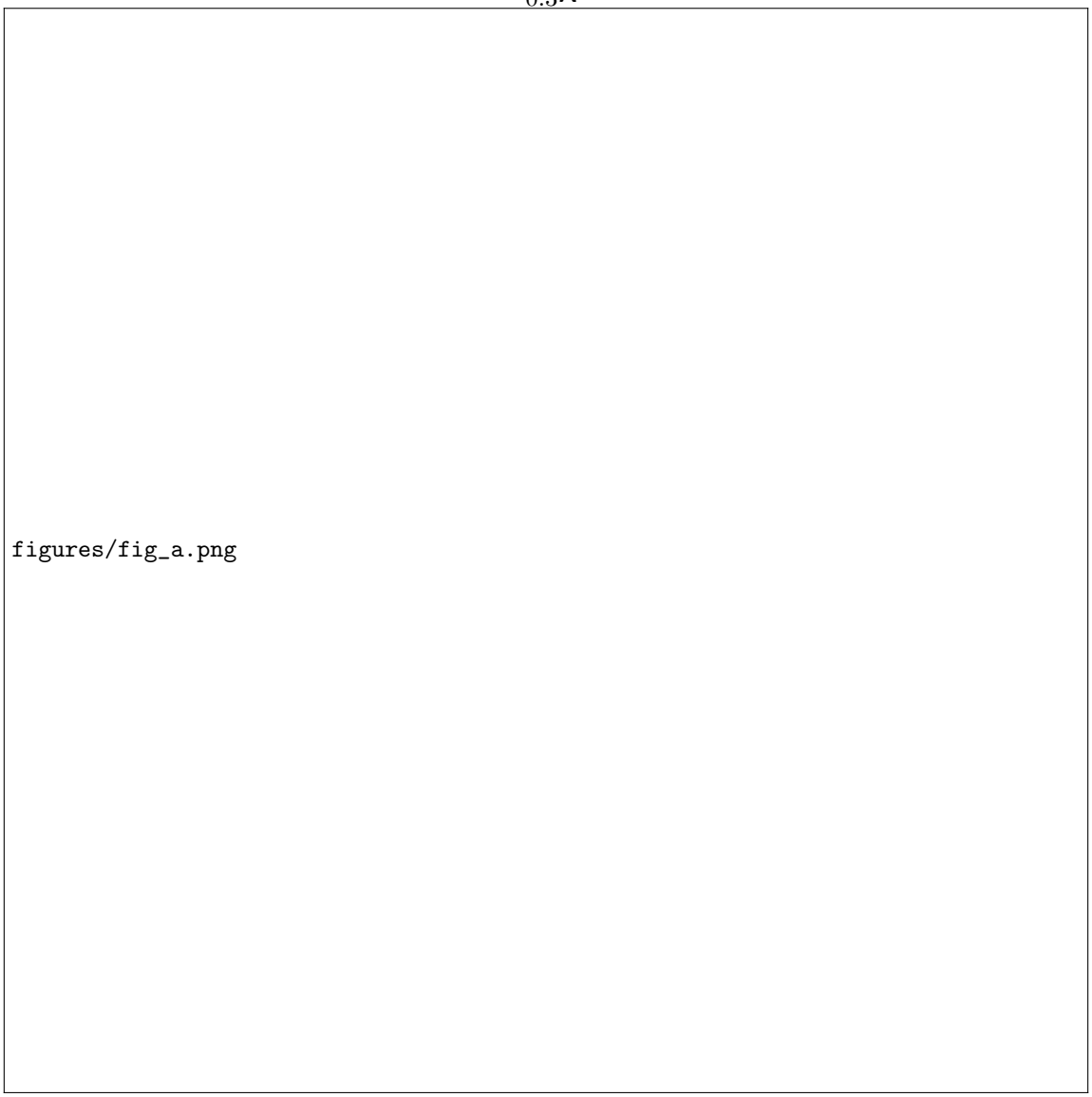
12 Some More Notation

As shown in Figure L1, the section number is inserted in the figure number. [13]
[14]

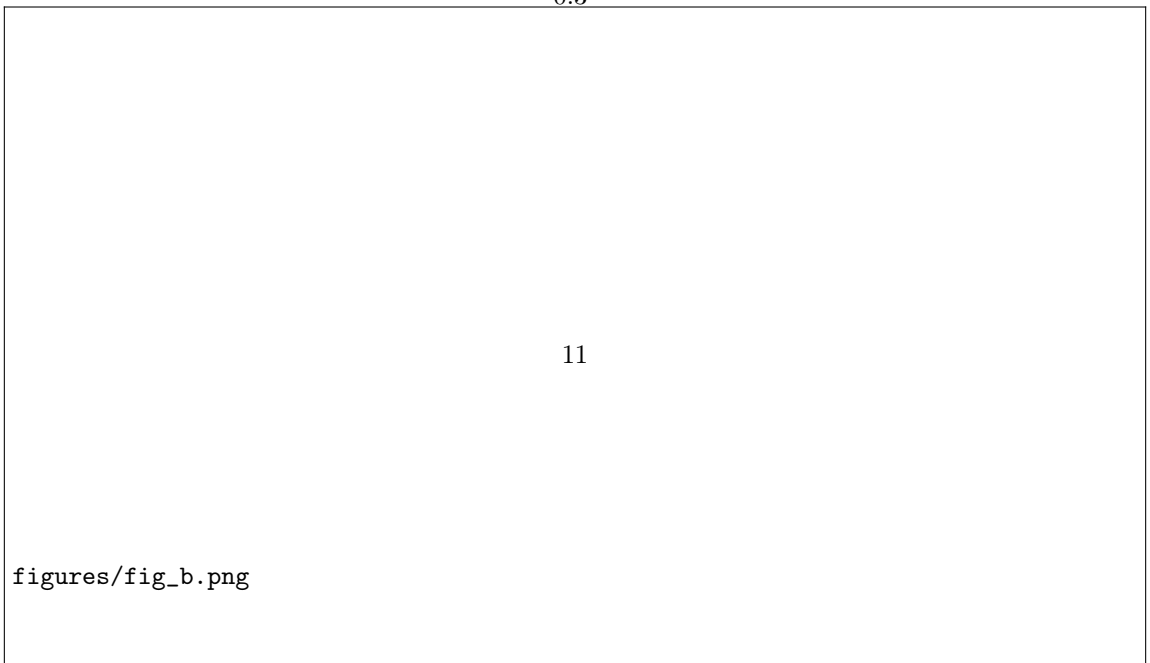
12.1 Appendix subsection title here

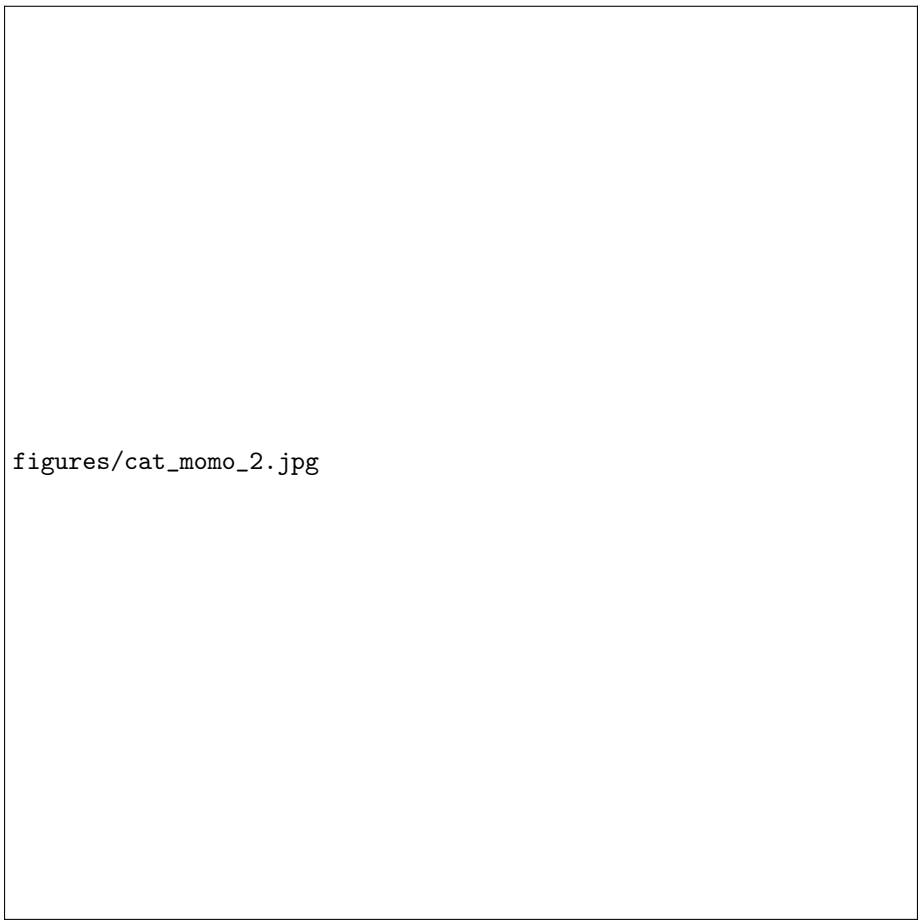
[15]

0.3**A**



0.3**B**





Gambar L1: This cat picture is located at the 'figures' folder.